

MAINS MASTER PROGRAM (MMP) 2024

ECONOMY-14; INFRASTRUCTURE-2

1. TABLE OF CONTENTS

1. Table of Contents.....	0	4) Outlook: Key steps being taken by Railways.....	15
2. Logistics Sector	1	5) IndiaN Railways Revenue CHallenge ...	15
1) Logistics Sector.....	1	6) Dedicated Freight Corridor (DFC)	16
2) National Logistics Policy	3	7) Railway Safety	17
3. Roadways	5	A) Kavach System.....	19
1) Roadways – Growth and Development	5	5. Water Transport.....	20
2) Rural Road Connectivity: Pradhan Mantri Gram Sadak Yojna (PMGSY)	7	1) Shipping and Port Sector.....	20
3) Investment Model for Road Sector	9	2) Maritime Amrit Kaal Vision 2047	22
4) Road Safety.....	11	3) PPP Model and Port Development	22
A) Bharat NCAP (Bharat New Car Assessment Program).....	13	4) Different Types of PPP Model Used in Port Sector	24
4. Railways	14	5) Coastal Shipping	26
1) Key stats	14	6) Ship Building.....	27
2) Key Challenges Faced by Railways	14	7) Ship Recycling	28
3) National Rail Plan Vision-2030.....	14	8) Inland Waterways.....	29

2. LOGISTICS SECTOR

1) LOGISTICS SECTOR

- Introduction

▫ **DEFINITION:**

- » Logistics, including transportation, inventory management, warehousing, material handling & packaging, and integration of information, is related to management of flow of goods between the point of origin and the point of consumption.

▫ **CONTEXT:**

- » As per the Logistics Performance Index, 2023 released by World Bank, India is ranked 38/139 countries in terms of the logistics performance.
- » Logistics Cost in India have been in the range of 14-18% of the GDP against the global benchmark of 8%.
- » Thus, logistic sector can play a crucial role in promoting the competitiveness of our industries. Besides it will also play a role in job creation and enhancing India's GDP.

▫ **DATA:**

- » India's logistic sector currently is worth \$160 billion dollars and government expects it to go to \$360 billion by 2032. But it remains unorganized to a large extent.

- Key Problems in India's Logistic Sector:

- **Complex Regulatory Environment:** Due to multiple regulations governed by various stakeholders. For instance, there are 20 government agencies, 37 export promotion councils, 500 necessary certifications; 200 shipping agencies; 36 logistic services etc.
- **Heavy dependency on Road Transport:** It is more energy intensive than other modes like railways and waterways.
 - In freight traffic roads have 66% share followed by railways (30%), Shipping (3%) and airways (1%).
- **Absence of seamless movement across modes:** This makes inter-mode movement a bottleneck.
- **Poor tech adoption** among various stakeholders: Government agencies still lack integrated IT infrastructure, automated approval etc.
- **Lack of qualified personnel** in supply chain.
- **High Indirect Costs** due to unpredictable supply chains and poor first and last mile connectivity add to the logistic cost.
- **High cost of logistics** - all the above reasons lead to logistics costing around 14-18% of the GDP.

- What should be done to improve logistics performance in India:

- **Streamline the regulatory framework:** Government should form a committee to understand how seamless logistical movement could be ensured by reduce regulatory bottlenecks.
 - » Enhance the competence and quality of logistics services - e.g. transport operators, custom brokers.
 - » Ensuring Efficiency of Clearance Process (i.e. speed, simplicity, and predictability of formalities) by border agencies, including customs
 - » Focus on timeliness in reaching destinations within the scheduled or expected delivery time.

- **Increase the share of railways and waterways**: The National Rail plan envisages the share to increase to 45% by 2030.
 - **Develop multi-modal hubs** which allow for seamless movement across different modes.
 - **Enhanced focus on tech-adoption**: IoT, AI, ML, Robotics etc can be used for efficient automation of the logistics sector.
 - **More International Collaboration**: Implement more initiatives like the India Middle-East-Europe Corridor (IMEC).
 - **Focus on Human Resource and Skill Development** to provide qualified personnel in supply chain.
 - **Prioritize sustainability** through energy efficiency of ships, other transport vehicles etc.
- **Key steps taken in this direction include:**
- i. **Process reforms** include - Introduction of GST, e-Sanchit, Single Window Interface for Trade (SWIFT), Indian Customs Electronic Data Interchange Gateway (ICEGATE), Turant Customs and others.
 - ii. **National Logistic Policy, 2022** was launched to integrate all government efforts by different agencies and lay on overarching inter-disciplinary, cross sectoral framework for logistics sector.
 - iii. **Infrastructure initiatives** like Bharatmala, Sagarmala, Parvatmala, National Rail Plan, UDAN etc.
 - iv. **Other initiatives improving the logistics sector include:**
 - **PM Gatishakti** brings together different stakeholders for integrated planning and coordinated implementation of infrastructure connectivity.
 - **35 Multimodal logistics parks** are being developed to enable seamless inter-modal transition.
 - **Creation of a Logistic Division** in the Department of Commerce
 - **Infrastructure Status to Logistics sector**
 - **Logistic Efficiency Enhancement Program and Vehicle Fleet Enhancement Program**
 - By Ministry of Road Transport and Highway
 - v. **Freight Smart Cities** by Commerce Ministry's Logistic Division - focuses on improving city freight movement.
 - vi. **Logistics Ease Across Different States (LEADS)** - an indigenous LPI to promote competition among states to improve the sector.

2) NATIONAL LOGISTICS POLICY

- **Practice Question:**
 - What are the targets of National Logistics Policy, 2022? Discuss the progress made under the policy so far [10 marks, 150 words]
 - "The National Logistic Policy 2022 has the potential to impact all three sectors of the economy - primary, secondary and tertiary" Discuss [10 marks, 150 words]
- The **vision of NLP** is to develop a technologically enabled, integrated, cost effective and sustainable logistic ecosystem in the country for accelerated and inclusive growth. It has set following targets:
 - **Reducing the cost of logistics** in India and to make it comparable to global benchmarks by 2030;
 - **Improve LPI ranking of India** - strive to achieve top 25 ranking by 2030
 - **Create a data driven decision support mechanism** for an efficient logistic ecosystem.
- The policy thus seeks to pave the way for India to become a logistic hub by leveraging technology, processes and skilled manpower.
- The Policy has **four key features (four key pillars)**: Integration of Digital System (IDS); Unified Logistics Interface Platform (ULIP); Ease of Logistics (ELOG); and System Improvement Group (SIG);
 - **Under IDS**, 30 different systems of seven departments are integrated
 - **ULIP** would bring all digital services related to the transportation sector into a single platform.
 - **Ease of Logistics (E-Logs)** is a new digital platform which has been started for industry associations to resolve issues by reaching out to the government.
 - **Systematic Improvement Group (SIG)** has been created along with the **Network Planning Group (NPG)** to improve coordination - between various ministries and between centre and state.
- The policy will be implemented through **CLAP (Comprehensive Logistics Action Plan)**, which proposes the following interventions:
 - **Integrating digital logistics systems** to develop a system of unified logistic interface
 - **Sectoral Plan For Efficient Logistics**
 - **Facilitating the development of logistics Park**
 - **EXIM logistics**
 - **Logistics manpower skill development and capacity building**
 - **Service Improvement Program**
 - **Standardizations of physical assets and benchmarking service quality standards**
 - **Engagement with different Indian states**
- **Progress (ESI 2023-24)**
 - **Integrated digital logistics system:**
 - Unified Logistics Integrated Platform (ULIP) has been developed.
 - For tracking 100% of India's containerized EXIM cargo, a Logistics Data Bank has been developed which uses RFID, IoT, etc. and is integrated with 29 port terminals of India, various toll plazas, container freight stations etc.

- **Sectoral Plans for efficient logistics:** Coal logistics plan and Policy was launched in Feb 2024. Comprehensive Port Connectivity Plan was prepared in 2022 that identified 107 port projects to strengthen connectivity between ports, railways, roadways, and inland waterways.
- **Facilitating the development of logistics Park:** Guidelines for multi-modal logistics park are being reviewed.
- **EXIM Logistics:** To streamline EXIM logistics, infrastructure gaps are addressed through action plans developed by the National Committee on Trade Facilitations
- **Capacity Building:** Training courses on logistics and PMGS-NMP are being integrated with central and state training institutes.
- **Service Improvement Framework:** The service improvement group has been established with the involvement of 30 business associations.
- **Engagement with States:** States are developing State Logistics Plans aligned with NLP to give policy focus at the state level. 26 states have notified their logistics policy.
- **Significance:**
 - NLP has the potential to enhance logistic sector by boosting effectiveness of the supply chain.
 - It would increase warehousing capacity and facilitate quicker movement of goods closer to consumer.
 - **Benefit other sectors** like manufacturing tourism etc.
 - **Improve productivity** of the workforce by reducing the time spent on travel
 - **Attract Foreign Investment** - efficient logistics is one of the key inputs used by foreign investors in decision making.
 - **Benefit farmers and rural areas** - reduce the transportation losses, improves access to market for farmers.
 - **Improve competitiveness** of India exports.
 - **Reduce regulatory complexity** by reducing multiplicity of authorities through ULIP.
- **Way Forward:** (Discussed in above topic)
- **Conclusion:**
 - The National Logistics Policy thus aims to bring ease of doing business in the logistics sector by simplifying the logistics processes, improving its efficiency bringing in transparency and visibility, and reducing logistics cost and time.

3. ROADWAYS

1) ROADWAYS – GROWTH AND DEVELOPMENT

- **News:** In Aug 2024, CCEA chaired by PM Modi approved 8 national high speed corridor projects under which 936 km length of highways will be constructed at the cost of Rs 50,655 crores
- **Introduction:** India has 2nd largest road network in the world with 66 lakhs kms of rural-urban roads and national state highways. The sector handles around 67% of countrywide freight and around 88% of passenger traffic annually.
- **Types of Roads:**
 - » India's road infrastructure is categorized into six categories:
 - **National Highways:** 1.32 lakh kms
 - » These are the primary roads of the country and connect large cities and big industrial areas.
 - » Their development and maintenance are the responsibility of central government.
 - » **Further need of development of National Highways:**
 - Out of the total roads, **only 2.06% is national highway**, but it's carrying capacity is 40%.
 - Higher the density of National Highway -> Higher the inter-state trade (Export + Import) as percent of GSDP (ESI)
 - A positive relationship exists between density of NHs and the per capita income in Indian States.
 - Presently, more than 70% of NHs are either **two lane or less**. Thus, there is a lot of scope of improvement.
 - **State Highways:** 1.79 lakh kms: These roads link all the important centre of industry, trade and commerce of the state and National Highways
 - **District Roads:** 6.12 lakh kms: These roads connect different parts of the district, important industrial centres and market centres and usually lead to local railways stations
 - **Rural Roads:** 45 lakh kms: These roads are found in villages and usually are of two types - Pucca (or metalled) and Kutcha (or non-metalled). They constitute 70% of roads in India.
 - **Urban Roads:** 5.41 lakh kms
 - **Project Roads:** 3.43 lakhs kms
- **Growth of Road Development in India:**
 - » The total road length in 1951 was 4 lakh kms; in 1991 it was 23 lakh kms and in 2019 it was 62 lakh kms.
 - » So, **in last 28 years (1991-2019)**, about 40 lakh kms of road has been added in the country.
- **Key reasons for improvement in road construction (both quality and quantity) in the last 3 decades:**
 - » Delinking Road Development and Direct Employment:

- Until the liberalization reforms (i.e. 1991), the road development was also connected with direct employment leading to labor intensive construction and also putting a cap on the quality of roads.
 - **After 1991**, the mindset changed and the use of **capital-intensive high-tech road making equipment was brought into use**.
- » **Better institutional framework:**
 - **National Highway Authority of India**, set up in 1995 increased the focus on quality of Highways.
 - **State-Level Road Development Corporations:** For e.g. Maharashtra was first state to set up the Maharashtra State Road Development Corporation Limited (MSRDCL) in Aug 1996. It developed Pune-Mumbai expressway.
 - **Creation of Focused Organizations:**
 - Indian Highway Management Company Limited (IHMCL) was set up to carry out electronic tolling.
 - National Highways & Infrastructure Development Corporation Limited (NHIDCL) was formed to develop roads in border states.
 - National Highways Logistic Management Limited was set up in 2020 for developing Multi-Modal Logistic Parks (MMLPs) and the first/last mile port connectivity projects.
- » **Higher capital investment:**
 - Capital investment by government and private sector rose from 0.4% in FY15 to about 1.0% in FY24 (around 3.01 lakh crores)
 - Infact, FY24 saw the highest private investment in the sector ever.
 - Funds garnered through **Asset Monetization** in the road sector have exceeded Rs 1 lakh crore since FY19.
- » **Important Projects** like NHDP (including Golden Quadrilateral and the North-South & East-West Corridors); Bharatmala Pariyojna etc.
 - **PMGSY** (focus on rural roads) success can be attributed to three reasons:
 - **Parvatmala Pariyojna** is focused on boosting last mile religious and tourist connectivity. Under it, six ropeway projects have been awarded.
- » **Increased focus on Expressways:** The first access-controlled expressways for fast and streamlined movement was opened between Mumbai and Pune in 2002. While the construction of expressway had a slow start, it has picked up in last 10 years. As of Aug 2023, India has about 5,000 km of operational express way, and other 9,000 kms under construction.
- » **Improvement in Road Making Technology:** As the NHDP rolled out, the import of road infrastructure was brought under Open General License, making it easy for procurement. Further, using the transfer of technology mechanism, domestic manufacturing was encouraged.
- » **Digitalization of Toll:** It has reduced the waiting time on toll from 734 seconds in 2014 to 47 seconds in FY24. Free flow tolling through Automatic Number Plate Recognition/ Global Navigation Satellite System has also been initiated.
- » **Focus on Sustainability:** E.g. 13.79 lakh tonnes of inert material from landfills have been used in urban extension road-II and Spur of the Delhi-Mumbai Express ways. **Recycling of bitumen & asphalt** is done during the brownfield upgradation of NHs.

- **Some key challenges which continue and need to be worked upon:**

- **Continuous Ribbon Development along the developed NHs** is posing a challenge for the construction of new parallel road/bypass. To solve this government is focusing upon Access Control Highways.
- **Land Acquisition** is another major issue which is hindering road development. Fast digitalization of land records can contribute to reducing the land acquisition delays.
- **Delay in environmental clearance** is also hindering several projects.
- **Safety:** India roads record more than 1.5 lakh deaths every year making Indian roads the most deadly in the world.
 - There is a need to improve engineering, promote awareness among drivers to avoid rash driving; ensuring buffer lanes for turns; and providing for sufficient roadside parking.
- **Congestion: Urban Roads (i.e. the city roads)** suffer from a lot of congestion in almost all cities and hampers first/last mile connectivity.
- **Lane Kms vs Road Kms:** As more multiple lane roads get constructed, it's important to measure lane kms. This will not only help us measure length but also the capacity.
- **Origin to Destination (OD) Data:** For future planning and development of road network, it is important to get OD data. This can also be used for ETC.
- **Better Coordination with PPP players:**
 - Significant time and energy are wasted in disputes between the PPP players and the authority. Projects get delayed leading to significant inconvenience.

- **Conclusion:**

- » Given the traction that India has built on road infrastructure, we should hope that the challenges are addressed, and the momentum enhanced for development.

2) RURAL ROAD CONNECTIVITY: PRADHAN MANTRI GRAM SADAK YOJNA (PMGSY)

- **Introduction:** Rural connectivity is a key component of rural development in India. The construction of rural roads brings various socio-economic benefits to the rural areas and result in forming a strong backbone for the agro-based economy.
- **Positive Impacts of better road connectivity in rural areas are multidimensional.** It results in diversified livelihoods' portfolio and improved quality of life for rural communities
 - a. **Agriculture:** Increase access to quality of inputs (seeds, fertilizer etc.) and market.
 - b. **Promotes local industries:** Roads lead to better access to raw material and market for the sale of finished products.
 - c. **Improved Employment opportunities:** Not only due to better agri-productivity and local industries, but also because of access to employment opportunities outside the village.
 - d. **Improved Health Access:** Due to easy access to health centres which becomes more crucial during emergency situation
 - e. **Improved educational opportunities** - Easy accessibility for schools and higher education facilities;
 - f. **Better law and order:** Improved mobility of police and armed forces; It also creates more deterrence in criminals.

- g. **Better implementation of government schemes/programs:** For e.g. Pradhan Mantri Ujjwala Yojana which aims to provide LPG cylinders to BPL families will be more effective in those rural areas which are better connected.
- h. **Enhanced Disaster Management**
 - Relief and rescue reaches more effectively
 - Rehabilitation work can also be faster in better connected areas.

Pradhan Mantri Gram Sadak Yojana (PMGSY)

- » PMGSY was launched in the year 2000, as a centrally sponsored program and a one-time special intervention by Ministry of Rural Development (**MoRD**).

Targets

- » PMGSY Phase 1: Habitation (250+) covered by March 2019.
- » PMGSY Phase 2 (Habitation 100+) and LWE blocks covered by March 2020.
 - Note: In 2016, under PMGSY a “Road Connectivity Project for LWE Affected Areas” was launched.
- » PMGSY Phase 3 focuses on sustainable maintenance of the roads.
 - Phase 3 was launched in July 2019
 - It is aimed at **consolidation of 1 lakh 25 thousand Kms** through routes and major rural links that connect habitations to GrAMs, Higher Secondary Schools, and Hospitals.

Progress (as of June 2024)

- » A total of 8,29,409 km of road length has been sanctioned under PMGSY out of which 7,63,308 of road length has been completed as on 18th June 2024 under various verticals at an expenditure of Rs 3.23 lakh crores (including state share)
- » 99.6% of the targeted habitations under PMGSY-I have been provided connectivity.

Key challenges being faced by PMGSY

- » **Adequate Maintenance** of the existing 4.6 million km of road network is emerging as a major challenge.
 - Many parts of the existing road network is either vulnerable or have already suffered damage from climate induced events such as floods, high rainfall, sudden cloud bursts and land-slides.
 - Generating funds for road maintenance has become a crucial challenge.
- » **Wide variation in unit costs among states** leads to confusions and delays
- » **Delays in execution**
 - Adverse weather condition
 - Issues in land acquisition
 - Law and order problems
- » **Non-availability of labor and material** especially in remote areas
- » **Lack of technically trained staff** to execute and monitor road work
- » **Providing public transport** in these roads.

Way forward

- » Climate vulnerability assessment during the design process to identify the critical locations affected by floods, water-logging, submergence, cloud bursts, storms, landslides, poor drainage etc.

- » **Special treatment for flood affected areas** through adequate waterways and submersible roads. Use of environmentally optimized road designs.
- » **For reducing the cost**, the use of non-traditional but durable and less expensive material such as iron and steel slag, fly ash and lime that are locally and easily available may be adopted.
- » **Proper Scheduling**: Schedule work after monsoon to avoid delays and cost over-runs
- » **Build Operate and Transfer Model** can be considered to deal with the issue of lack of resources.
- » **Better Monitoring** can be achieved by giving the responsibility of monitoring and undertaking minor maintenance to Gram Panchayat.
- » **Skill development** of rural workforce should also focus on road construction and maintenance activities.

3) INVESTMENT MODEL FOR ROAD SECTOR

- **News:** CCEA approved eight high speed corridor projects under which 936-km length highways will be constructed at a cost of Rs 50,655 crores. (Aug 2024)
- **Significance:**
 - » Improved road infra and associated advantages
 - » 4.2 crore mandays of direct and indirect employment.
- **Background**
 - » The erstwhile Planning Commission developed the first version of Model Concession Agreement (MCA) for PPP in highways in 2006. This was done considering the need to standardize documents and process for the PPP framework in the country for ensuring uniformity, transparency and quality in development of large-scale infrastructure projects including highways.
 - » Subsequently, the Planning Commission has developed several other versions of the MCA for highways considering the different PPP modes like BOT (Toll), BOT (Annuity) and EPC addressing to a significant extent the changing need of the sector.
- **Engineering, Procurement and Construction (EPC)**
 - » Under this format, the government authority that owns the project hires one or more private companies (or another public company) to construction the project.
 - The private players bid for the project and build it within pre-specified time and cost. It doesn't have any role in the road's ownership, toll collection, or maintenance.
 - Government with full ownership of the road, takes care of the toll collection and maintenance of the road.
 - » Before the arrival of other PPP model, projects were built through EPC model.
 - » **Risk to companies?**
 - Almost no risk, barring road construction risk. So margins were also low.
 - » **Limitations**
 - **High Financial burden on government:** EPC was the preferred mode of highway project development in 2013-14 and 2014-15, it suffered from an inherent limitation - the financial resource available with the government.
 - » **How much is this model being used?**
 - One of the most common method still being used by government for construction of National Highways

- **Build Operate and Transfer Model (BOT)** : In BOT the developer invests in the project. Here the private player owns the asset and recovers his investment through toll charges or collect a fixed annuity from the government for a contractually pre-specified period of time (typically 15 to 30 years).

A. **BOT (Toll)**

- » The private player is also responsible for operation and maintenance once the project is commissioned.
- » **Risk to private players**: Own traffic risk
- » **Number of takers are coming down** since 2012 due to associated risk involved.
- » **Revision of BOT-Toll Concession Model (March 2024)**
 - More liberal norms for compensation, extension of concession period, and terminal payments.
 - A new concept called "**Construction Support**" has been brought under which NHAI pays 40% of the total project cost to the company building the highway in 10 installments linked to physical progress in construction. Earlier, NHAI only provided the equity support to the company.

A. **BOT (Annuity)**

- » Here the contracting authority (government) pays the pre-specified sum of money semi-annually or annually to the private player and the **traffic risk (toll revenue risk) is completely borne by the government**(contracting authority).

B. **Hybrid Annuity Model (HAM)**

- » Approved by Cabinet in Jan 2016
- » **Main Provisions**
 - It is a mix of EPC models (40%) and BOT(Annuity) (60%).
 - This is the **fourth PPP model** (BOT(Toll), BOT(Annuity) and EPC) to be introduced in India for the execution of road projects and is intended to kick start stalled projects and accelerate highway construction.
 - Under this PPP model, the government invests 40% of the construction cost for building highways over a period and balance comes from the private developer. The government invest money in five equal instalments based on the targeted completion of the road project.
 - Bidding rolls out only after 90% land is available.
 - The government pays a semi annuity/annuity premium to the developer, thus shouldering the traffic risk.
 - A. Assured returns and insulation from revenue/traffic risk for private players makes the model attractive.
- » **Main objective** is to revive highway projects in the country by making one more mode of delivery of Highway Projects.

- **Advantages**

- Revival of the projects
- Attracting new contractors and return of old ones
- Stress Free Lending
- Speedy Completion of Project

- **Conclusion**

- Considering India's large infrastructure investment requirements, only the combination of resources of both private and public sector can ensure fulfilling of financial requirement. In developing countries, the contribution of private sector is still very low and successful PPP models can play a very significant role in enhancing this contribution.

4) ROAD SAFETY

- Why in news?

» Road accident statistics by MoRTH (released in Nov 2023)

- In the year 2022, India saw a total of 4,61,312 road accidents and lost 1.68 lakh lives (increase of 9% from last year)

- Introduction

- Safe roads are of paramount importance for the growth and development of any country. But as per the latest numbers by MoRTH, in 2022, India saw a total of 4,61,312 road accidents and 1.68 lakh fatalities due to these accidents.

- The severity of road crash (measured by the number of people killed per 100 accidents), has increased over the past decade from 28.2% in 2012 to 36.5% in 2022, with a consistent increase every year.
- According to Geneva-based World Road Federation's World Road Statistics, **India is most unsafe country in the world** followed by China (63,000 deaths) and the U.S.A (37,000 deaths)

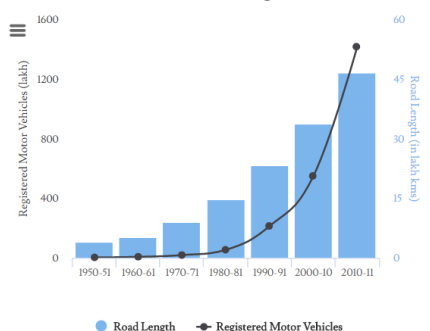
- Other key facts about Road Safety in India:

- National and State Highways which account for only 4.9% of the total road network, witnessed 56.1% of all accidents and almost 61% of fatalities.
- Most important causes of road accidents were over-speeding, wrong side driving, drunken driving, use of mobile phones etc.
- As per government estimates, road accidents cause a loss of around 3% of country's GDP (which is double of loss by developed countries).
- More road accidents also lead to increased premium of third-party insurance.

- Key factors behind high number of road accidents and increasing severity of road accidents:

- Violation of traffic rules: As per MoRTH biggest cause is rash driving, wrong side driving, mobile phone usage etc.
- Increasing congestion on Indian roads: The rise in number of vehicles have outpaced increase in road length.
- Quality of engineering of roads -> presence of several black spots on Indian roads.
- Quality of Vehicles: Most of the vehicles in India fail the safety tests under European standards.
- Improper traffic management - especially in tier-2 and tier-3 towns
- Harassment of good Samaritans -> This reduces the help during the golden hour leading to increased severity.

Rise in number of vehicles has outpaced increase in road length



- **Recent MoRTH have taken several steps** to improve road safety:
 - **Strengthening of laws and proper implementation:** MVA Amendment 2019 has led to higher fines, more insurance, focus on golden hour treatment and reduced harassment of good Samaritans.
 - **Increased focus on road Safety:**
 - National Road Safety Policy focuses on increasing awareness, encouraging safer road infrastructure, better enforcement of laws, trauma care etc.
 - Road safety has been made integral part of road design at planning stage.
 - Road Safety Audit of selected stretches of NHs has been taken up.
 - High priority has been accorded to identification and rectification of **black spots** (accident prone spots) on National Highways. At total of 789 such black spots have been identified for improvement.
 - **India Road Assessment Program (IndiaRAP)** launched by MoRTH in Nov 2007. The aim is to tackle high risk roads around the country.
 - **4-E Strategy:** the strategy by MoRTH focuses upon Education, Engineering (both roads and vehicles), Enforcement and Emergency care. Based on this draft action plan has been shared with states.
 - **Better Training:** Setting up of model driving training institutes in states and refresher training to drivers of Heavy Motor Vehicle in the unorganized sector.
 - **Focus on better safety features in vehicles:** **Bharat New Car Assessment Program (Bharat NCAP)** has been launched as an indigenous star-rating system for vehicles rolled out by MoRTH.
- **Way Forward:**
 - i. **Emphasise on stricter implementation of traffic norms:** As most of the road accidents are taking place because of issues like rash driving, mobile phone use etc., stricter implementation of traffic rules can increase road safety drastically.
 - Increase penetration of electronic monitoring of violation of traffics and proper imposition and collection of fines by the violators.
 - Human resource in the road traffic department also needs to increase for proper implementation of all the norms.
 - ii. **Awareness Generation:** There is a need of mission mode nationwide campaign like **Surakshit Bharat Abhiyan** to bring a cultural change towards safe driving in the country.
 - iii. **Improvement of Road Design:**
 - Increased road safety audits for new projects (especially in the EPC project) and in the black spot areas.
 - Create new institutional structure which will generate new knowledge and thus new road standards.
 - iv. **Improvement in traffic management:** Especially in small towns through installations of traffic signals, signage etc.
 - v. **Emphasize on Public Transport:** This will reduce road congestion and will contribute towards road safety.
 - vi. **Need of improved trauma care:** To reduce the severity of the accidents (which are continuously increasing currently)
- **Conclusion:** An holistic and integrated approach which includes following of rules by travellers, imposition of penalties by authorities, development of good quality of roads by government and

development of good qualities of vehicles by manufacturers will be needed to enhance road accidents and reduce the severity of road accidents in the country.

A) BHARAT NCAP (BHARAT NEW CAR ASSESSMENT PROGRAM)

- In 2023, the MoRT&H have rolled out an **indigenous star-rating system** for crash testing. It is a voluntary rating system and came into effect from 1st Oct 2023.
- **Key features:**
 - » Applicable to passenger vehicles with not more than eight seats in addition to driver's seat with gross vehicle weight not exceeding 3,500 kg.
 - Only the base model of the particular variant will be tested.
 - » Cars voluntarily nominated by automobile manufacturers will be crash tested as per the protocols laid down in the Automotive Industry Standard 197.
 - **Note:** The protocol adopted by Bharat NCAP is based on Global NCAP, which is a project of the UK-based NGO, Towards Zero Foundation.
 - » Cars will be assigned a rating between one star to five star after being evaluated on three parameters - Adult occupant protection; Child Occupant protection; and safety assist technologies present in the car.
 - » Vehicles are assigned between 1 to 5 stars indicating their safety to collision.
 - » Though Bharat NCAP is voluntary, in certain cases, cars may be subjected to a crash test such as for a base model of a popular variant (minimum clocked sale of 30,000 units), or when the MoRT&H recommend a model for testing based on market feedback or in the interest of public safety.
- **Significance:**
 - » Help customers take informed decision while purchasing a car.
 - » Increase demand for safer cars -> encourage manufacturers to increase safety of vehicles.
 - » **Reduced cost of testing:** It will cost around Rs 60 lakh to car manufacturers (in case of overseas test the cost was around 2.5 lakh).
- **Way forward:**
 - » **Upgradation of Protocol:** Align Bharat NCAP with global standards by expanding testing parameters. For e.g, the US NCAP also include a roll over test which tests whether a vehicle is vulnerable to tipping up on the road in a severe manoeuvre. JAPAN's NCAP covers electric shock protection after a collision, performance of neck injury protection in a rear-end collision, passenger seat belt reminder evaluation, evaluation of pedestrian protection technologies etc.

4. RAILWAYS

1) KEY STATS

- Railways in India began in **1853**, when the first train steamed off from **Mumbai to Thane**, a distance of **34 kms**.
- **Current capabilities (ESI 2023-24)**
 - » Indian railways is **the fourth largest railway network** in the world (after US, Russia and China).
 - » It **has route length of 68,000 kms** and **around 1,30,000 bridges**.
 - » IR include **7,335 stations** and serves **2.4 crore passengers daily (largest)** and **204 million tonnes** (4th largest) of freight daily.
 - » It is also **the largest employer** in the country **with 12.54 lakh** employees (as of 1st April 2024)
- **CAPEX:**
 - » Capex has **increased by 77% over the last 5 years** (Rs 2.62 lakh crores in FY24) with **significant investments in the construction of new lines, gauge conversion, and doubling**.
- **Budget 2024-25:**
 - » The **Ministry of Railways** have **received a record allocation of over Rs 2.55 lakh crore** in FY25 which is a jump of **5.85%** from the previous year's Rs 2.41 lakh crores.
 - » **Total outlay for capital expenditure** is **Rs 2.65 lakh crore** (Rs 2.52 lakh crore from general revenue, Rs 200 crore from Nirbhaya Fund, Rs 3,000 crore from internal resources, and Rs 10,00 crores from extra budgetary resources).

2) KEY CHALLENGES FACED BY RAILWAYS

- **Ageing Infrastructure**
- **Losses in passenger segment** -> this leads to cross subsidization from cargo segment
- **Resurge in accidents recently**
- **Lack of participation of private players:**
- **Challenges in Cargo segment:**
 - **Decreasing modal share**
 - **Heavy dependence on transportation of a few items** (for e.g. coal)
 - **Poor terminal facilities** -> leads to delays in loading and unloading.
- **Revenue challenges** -> reduces investment and hampers the quality of services.

3) NATIONAL RAIL PLAN VISION-2030

- Indian railways have prepared a National Rail Plan (NRP) for India - 2030.
- The plan is to create **future ready Railway system by 2030**.
- The NRP is aimed to **formulate strategies based on both operational capacities and commercial policy initiatives** to **increase modal share of the Railways in freight to 45%** (at present it is around 27%) and to sustain it.
- **Other aspects:**
 - » **Reduce transit time of freight** substantially by **increasing average speed of freight trains to 50Kmph**.

- » As part of the National Rail Plan, **Vision 2024 has been launched** for accelerated implementation of certain critical projects by 2024 such as: 100% electrification, multi-tracking of congested routes, upgradation of speed to 160 kmph on Delhi-Howrah and Delhi-Mumbai routes, upgradation of speed to 130kmph on all other Golden Quadrilateral-Golden Diagonal (GQ/GD) routes and elimination of all Level Crossings on all GQ/GD route.
- » Identify new Dedicated Freight Corridors and new High Speed Rail Corridors.
- » Assess rolling stock requirement for passenger traffic as well as wagon requirement for freight.
- » Assess Locomotive requirement to meet twin objectives of 100% electrification (Green Energy) and increasing freight modal share.
- » **Assess the total investment in capital** that would be required along with a periodical break up.
- » Sustained involvement of the Private Sector in areas like operations and ownership of rolling stock, development of freight and passenger terminals, development/operations of track infrastructure etc.

4) OUTLOOK: KEY STEPS BEING TAKEN BY RAILWAYS

The key focus areas for Indian Railways is fast capacity augmentation, modernization of rolling stock and maintenance, improving quality of service and energy efficiency. Keeping this in mind, **investments are prioritized in:**

- i. **Dedicated Freight Corridor (DFC)**
- ii. **High Speed Rail**
- iii. **Modern Passenger services** like Vande Bharat, Amrit Bharat Express, Aastha Special Trains, High Capacity Rolling Stocks, and last mile rail linkages.
- iv. **Projects for three major corridors:** (1) High Traffic Density Corridors; (2) Energy, Mineral and Cement Corridors (3) Rail Sagar (port connectivity) corridors are also planned to reduce the logistics cost and carbon footprint.
- v. **For reducing carbon footprint** railway is focusing on sourcing energy from renewable sources. The expected requirement of installation of renewable capacity by 2029-30 is around 30 GWs. Other strategies include shifting from Diesel to Electric traction, promotion of energy efficiency, and afforestation.

5) INDIAN RAILWAYS REVENUE CHALLENGE

- **Key Challenge:**
 - » **High Operating Ratio:** While the Indian Railway's capital expenditure has increased in the recent years, IR's **Operating Ratio**, which is the ratio of ordinary working expenses to the gross traffic receipts, has shown no improvement.
 - » **Trap of Rising Debt:** Lack of surplus, leads to the need of augmentation of funds from Gross Budgetary Support (GBS) and Extra Budgetary Resources (EBS). The EBS leads to the need of repayment of principle and interest.
- **But, it is important to continue to invest in railways** as investment in railways boost manufacturing and services, tax revenue for government and allows for more job opportunities.
- **Where is the problem?**
 - » IR's freight segment is profitable, whereas the passenger segment is making huge losses.

- For e.g. CAG report presented to Parliament in Aug 2023 states that there was a loss of Rs 68,269 crores in all the classes of passenger services during FY22, with all profit from freight traffic nullified in cross subsidizing passenger services.
- » **Political Economy** makes it difficult to increase the passenger fares substantially. So, **increasing profits from freight traffic** by increasing freight volume is crucial.
- **Indian Railways Decreasing Share in Freight Traffic:**
 - » IR's modal share in India's freight traffic has steadily decreased from 80% at the time of independence to 27% currently.
 - » **Why the decrease?**
 - **Competition from other modes:**
 - Improvement in roadways infrastructure, and successive price hikes in railways etc.
 - Pipelines and coastal shipping have also forayed into transportation of bulk items.
 - **Poor Performance in non-bulk freight transport**
 - For e.g. in parcel segment, the tariff is pretty high. After adding the first and last mile costs, the prices are higher than truck rates.
 - Other challenges are improper terminals, inconsistent weighbridges, unreliable transit times, complex booking and delivery mechanisms.
 - **Lack of Diversity:** 11 commodities in the IR's transport basket account for 90% of tonnage and revenue, of which coal is around 45% and iron and cement 10% each.
 - **Slow average speeds** of freight trains of around 25 km/hour.
 - **Lack of participation of private players:** It should be noted that private container train operation policy, initiated in 2006 to boost the rail share of container movement, has not made any significant dent in improving the share.
 - » **Recent Steps taken:**
 - **Dedicated Freight Corridor (DFCs)**
 - **Gati Shakti Cargo Terminals**
 - **Focus on expanding freight basket** - by targeting more commodities (e.g. Kishan Rail)
 - Policy of "**Long term Tariff Contracts**" with major customers
 - Double Stack dwarf Container train under a wire - a new delivery model
 - **RO-RO Services** - to reduce road congestion and environmental improvement.
 - » **Way Forward:**
 - **Rationalize the train fares** in a phased manner for passenger traffic.
 - **Increasing General Cargo Tonnage:** By attaching **two or three parcel vans in all popular trains**, ensuring their timely movement and tweaking tariff rules, the Indian Railways should be able to load substantial general cargo tonnage in the coming years.
 - **Develop common user facility at cargo aggregation and dispersal points** in mining clusters, industrial clusters, and large cities.
 - Since the knowledge of these clusters rest with states, collaboration with state governments is of utmost importance.
 - **Encourage and liberalize the design of new wagons amenable to higher and efficient loading to deal with new commodities.**

6) DEDICATED FREIGHT CORRIDOR (DFC)

- **Dedicated Freight Corridors (DFCs)** are high speed and high-capacity railway corridors that are exclusively meant for the transportation of freight or goods commodities.
- India's Ministry of Railways has undertaken the construction of **two DFCs** namely:
 - i. **Eastern Dedicated Freight Corridor:**
 - **1,840 kms** (between Ludhiana (Punjab) to Sonnagar (Near Kolkata, WB)). It will also include PPP section of the Sonnagar-Dankuni route.
 - It will run through Delhi.
 - States covered include Punjab, Haryana, Uttar Pradesh, Jharkhand, Bihar and West Bengal.
 - ii. **Western Dedicated Freight Corridor** (between Dadri (Uttar Pradesh) and Mumbai (Jawaharlal Nehru Port Terminal (JNPT))).
 - **1,506 kms**
 - States covered: **UP, Haryana, RAJ, Gujarat, and MHA**.
- The DFCs are being developed by **Dedicated Freight Corridor Corporation of India Ltd (DFCCIL)**.
- The Funding for DFC is through **World Bank (US\$ 2.725 billion) for EDFC**, and **Japan International Cooperation Agency (JICA) loan (38,722) for WDFC** and rest from the Gross Budgetary Support (GBS).
- **Expected Impact**
 - » Improved infrastructure and capacity of railways.
 - » Reduce travel time on the two routes for both passengers and goods.
 - DFCCIL will run freight trains at the maximum speed of 100 kmph as against the current maximum speed of 75 kmph on Indian Railway tracks whereas the average speed of freight trains will also be increased from existing speed of 26 kmph on Indian Railway lines to 70 kmph on DFC.
 - » Benefit industries in the region by speeding up transportation.
 - » Reduce pollution as the railway transportation is more efficient.
- **Progress:**
 - » As of April 2024, India Railways have operationalized 100% of the eastern arm and 85% of the western one.
 - » **Average speed** of the trains is around 50-60 kmph (and can be increased upto 100 km per hour)
 - » **Estimated** cost of operationalizing the network currently stand at Rs 1,24,000 crores.

7) RAILWAY SAFETY

- **Introduction:** Several railway tragedies in recent past including the Balasore accident in June 2023 and the Darjeeling accident in June 2024 have highlighted the key challenges faced by Indian Railways in ensuring safe and accident-free travel and transportation.
- **Key factors making Indian Railways vulnerable:**
 - » **Infrastructure Challenges:**
 - **Ageing Infrastructure:** Portions of Indian railways' track, bridges, signaling system are pretty old and require upgradation.
 - **Inadequate track maintenance:** Leads to track defects and derailment.
 - **Unmanned level crossings:** Though railway has eliminated unmanned level crossings on its broad gauge trains, but they still exist on meter gauge and narrow gauge.
 - **Kavatch** - the anti-collision system covers less than 3% of the Indian railways currently.

- » **Human Resource related challenges:**
 - **Human errors** by train crew, station masters, and other personnel leading to signal violation, Overspeeding, and miscommunication. It also happens due to **staff being overburdened and overworked**.
 - » **Natural disasters and extreme weathers** may lead to floods, landslides and can pose safety risks for railways.
- Important Past Steps to Ensure Railway Safety:
- i. **High Level Safety Review Committee** chaired by Anil Kakodkar submitted its report in 2012 and recommended modernization of tracks; elimination of level crossings by building rail over and under bridges; strengthening of railway bridges; 100% mechanization of track maintenance.
 - ii. **Rashtriya Rail Sanraksha Kosh (RRSK)** was created with a corpus of Rs 1 lakh crore over a period of 5 years for giving a major boost to safety related works.
 - iii. **Induction of technology for safety improvements – Smart Coach:** These coaches have diagnostic system to monitor bearing vibrations and provide advance information on health of bearing wheel & track.
 - iv. **Complete switchover to LHB:** Indian Railways have decided to completely switch over to manufacture of LHB design main line coaches from 2018-19 onwards.
 - v. **Provision of Electronic Interlocking (EI):** To increase safety and flexibility EI is being adopted on large scale to derive benefits of digital technologies in train operation and enhance safety.
 - vi. **KAVACH Technology (Automatic Train Protection System)** is being installed in Indian Railways
 - vii. **Other steps include**
 - » Number of **stations with the installation of CCTV based camera surveillance** have increased to more than 500.
 - » **Indian Railway Institute of Disaster Management** have been opened up in Bangalore for training of officers and staff.
 - » **Commandos for Railway Security** - Launch of first Railway Commando battalion 'CORAS' on 14th Aug 2019 to tackle the menace of terrorism and Naxalism.
 - » **Empowerment of RPF** to make seizure under Narcotics, Drugs and Psychotropic Substances (NDPS) Act
 - GoI through a notification in April 2019, empowered RPF to make seizures and arrest under NDPS act.
 - Subsequently, RPF have recovered large number of such recoveries.
 - » **Operation AAHT:** Under Operation Aaht, RPF teams are deployed on all long distance trains/routes with focus on rescuing victims, particularly women and children, from the clutches of traffickers.
- **Way Forward:**
- **Infrastructure improvement:**
 - » Installation of KAVACH and Electronic Interlocking system should be taken up in mission mode.
 - As per ESI 2023-24, Kavach is deployed in 2.14% of India's rail track coverage; and only 46% of the railway stations have made the switch from mechanical signaling to electronic interlocking system.
 - » Improving signaling system
 - **Human Resource Deployment Problem** specially in telecommunication-signaling wing.

- » At the ground level, it is saddled with semi-skilled workers promoted from linemen towards handling mechanical or sophisticated electronics system.
- » **Reforming Recruitment Process** - Highly qualified engineers from good institutes with better pay; Continuous training program; and counselling to ensure safety

A) KAVACH SYSTEM

- **KAVACH** has been developed indigenously by RDSO in association with three Indian vendors and it has been adopted as our National Automatic Train Protection (ATP) System.
 - It aids Loco Pilot to avoid Signal Passing At Danger (SPAD) and over speeding and also help in train running during inclement weather such as dense fog.
- **Key features:**
 - Controls speed of the train and automatically applies brake if the train is at the risk of overshooting a red signal.
 - Repeats line-side signal in cab which is very useful for higher speeds and foggy weather
 - Auto Whistling at LC gates
 - Collision avoidance by direct to loco communication
 - Supports feature of SOS in case of any mishap to control train in vicinity.

5. WATER TRANSPORT

1) SHIPPING AND PORT SECTOR

- **Introduction: Basic Information**
 - » For ages, the sea has been source of opportunity and seashores have acted as the gateway to prosperity for India. The development of ports and shipping sector is crucial for the economy given that most of the international trade is handled through ports (around 90% of international trade cargo by volume and 79.9% by value).
- **Shipping and Port Sector has a lot of potential in India:**
 - » **India's Geography:** India is endowed with a rich coastline of ~7500 km and has a strategic location on key international maritime trade routes.
 - » **India's fleet** is just 1.2% of the world's fleet and carries only 8% of India's Exim trade.
- **Why Port sector is significant for India?**
 - » **Economic development:** It contributes to reduction in logistics cost and thus increasing competence of Indian economy.
 - » **Enhances international trade** - and also ensures easy export opportunities for Indian businesses.
 - » **Sustainability:** Transportation through ships is way more energy efficient than transportation through railways and roadways.
 - » **Security:** Development of ports can enhance security of strategic installation and curtail transnational organized crime at sea.
 - » **Strengthening International relations:** Port sector can contribute to regional integration and counter China's influence in the region.
- **Key achievement of Shipping Sector in last 10 years (ESI 2023-24)**
 - » **Doubling of capacity of major ports:** From 871.5 million tonnes per annum at the end of March 2014 to around 1600 million tonnes in 2024.
 - » **Consistent rise in container traffic** from 555 million tones in FY14 to 796 million tonnes in FY24.
 - » **Rank in International Shipping category** of World Bank's LPI has improved to 22nd in 2023 from 44th in 2014.
 - » **Container turnaround time** has dropped by 50% since 2014
 - » **Increasing profitability** -> Operating ratio has come down to 48% (from 53% in 2020-21)
- **Some challenges faced by India's Shipping Sector:**
 - i. **Inadequate linkage of ports with hinterlands** - The evacuation of cargo is slow, leading to congestion.
 - ii. **Declining share of Indian ships** in the carriage of India's overseas trade (from about 40% in 1980s to 7% in 2015-16). This leads to a lot of forex outflow.
 - Ship owning, chartering, financing and building remain largely out of bounds for Indians. India hasn't simply moved up the shipping value chain.
 - iii. **Ageing Indian fleet:** The average age increased from 15 years in 1999 to 19.3 years as on Jan 1, 2017. In fact, around 45% of the fleet is more than 20 years of age.
 - iv. **Competition from neighbouring countries** like China and Sri Lanka is also impacting India's port and shipping sector.

- v. **Average turnaround time** still remains lower than advanced economies due to port inefficiencies.
- vi. **Resurgence of Piracy in 2023 and 2024** also hampers international trade using ships.

- **Steps Taken:**

- » **Sagarmala initiative**: Since 2015, under this several projects worth Rs 5.8 lakh crore have been undertaken across **five key areas** of Port Modernization & Fresh Development, connectivity enhancement, port led industrialization, coastal community development, and Coastal Shipping and Inland water transport.
- » **Major Port Authorities Act, 2021** focuses on decentralized decision making, professionalism, and PPP Models and has enhanced efficiency and improved governance of ports.
- » **Maritime Amrit Kaal Vision 2047** focuses on 11 overarching themes to transform India's maritime sector.
- » **Various steps are being taken to enhance efficiency of ports:**
 - Improving Port Governance; addressing low utilization; modernization of ports with new technologies; etc.
 - **Digitization of key EXIM processes** -> streamline compliances and reduce turn around time (TAT)
 - E.g. Port Community System (PCS 1x) has digitized e-invoice, e-payment etc.
- » **Promoting Indian Shipping Companies: Scheme to provide subsidy support to Indian shipping companies** (announced in July 2021) in global tenders to promote flagging of merchant ships in India
- » **Enhancing Sustainability: 'Harit Sagar' - Green Port Guidelines** - launched in May 2023: Under this 4 major ports are already producing more renewable energy than demand.
- » **Enhancing monitoring and evaluation: 'Sagar Ankalan'**, released in Feb 2024, is a national benchmarking of Indian ports performance applicable to all India seaports

- **Way Forward:**

- i. **Improved investments** for ensuring enough funds for MIV 2030
- ii. **Incentivize private investment** in port and shipping sector through tax breaks, subsidies and loans
- iii. **Domestic Ship Building Industry** needs to be encouraged to reduce the cost of ships.
 - Ship building and owning would give India a seat at the table not only in the global maritime industry but also enhance its presence in the international trade.
- iv. Focus on **niche areas** like coastal shipping and cruise shipping.
- v. **Regulatory reforms**: Ensure level playing fields for major and minor ports.
- vi. **Work towards sustainability**: The shipping sector is a major polluter, and there is a need to promote greener fuel and technology in shipping sector
- vii. **Promote Cooperation with neighbouring countries** - the government should promote cooperation with neighbouring countries in the shipping sector. This could include joint ventures, sharing of resources, and harmonization of regulations.

- **Conclusion:**

- » The ports and shipping sector is a vital part of the Indian economy. By taking further steps to develop the sector the government can help boost the economic growth and create jobs.

2) MARITIME AMRIT KAAL VISION 2047

- **The Amrit Kaal Vision 2047** is a long term development roadmap launched by the Indian government in 2023. It aimed to transform India into a developed nation by 2047, the 100th year of its independence.
- **Maritime Amrit Kaal Vision 2047** is a sub-vision under the larger plan. It focuses on developing India's maritime sector, with emphasis on:
 - i. **Expansion of Port Infrastructure**: Development of six mega ports with capacities exceeding 300 to 500 million tonnes per annum.
 - ii. **Sustainable growth**: By promotion of environment friendly practices and blue economy.
 - iii. **Technological Advancements**: Investment in R&D for advanced maritime technologies
 - iv. **Enhanced Connectivity**: Developing efficient port infrastructure and trade corridors to strengthen connections.
 - v. **Skilled Workforce**: Training and skill development program to cater to the evolving needs of the maritime sector.
- **It aims** to develop world class ports, promote inland water transport, coastal shipping, and a sustainable maritime sector.

3) PPP MODEL AND PORT DEVELOPMENT

- **Example Questions:**
 - "PPP projects in the ports sector are of great significance in making Indian ports more competitive and strengthening India's position in international trade" Discuss.
 - Evaluate the impact of the PPP model on the efficiency and performance of ports in India. Provide examples of successful PPP projects and their outcomes. [15 marks, 250 words]
- **What is PPP and different models of PPP** (Already done in the basic's class)
- **PPP Model in Port Sector**
 - It has been 27 years since the PPP model was first introduced in India's major ports. In July 1997, the Jawaharlal Nehru Port entered into first agreement with the private player Nhava-Sheva International Container terminal within the Jawaharlal Nehru Port
 - When the PPP started, there was nothing like a **Model Concession Agreement**. Government came up with various templates and through learning along the way, these agreements are being suitably modified to take care of the concerns of the government sector and private sector.
 - **Various PPP Models used in the port sector:**
 - Landlord Port Model
 - Build Operate Transfer (BOT)
 - Build Own Operate Transfer (BOOT)
 - Build Own Lease Transfer (BOLT)
- **How has PPP contributed over the years in port sector?**

- » **Problems:** Before PPP was introduced, the port sector was constrained by limited capacity, traditional infrastructure, and poor equipment levels. These were resolved by roping in the private sector to make ports globally competitive.
 - » **Improved Port Infrastructure:** PPP model has hastened the economic growth by introducing international level infrastructure in the port. It has enabled national ports to compete with international ones.
 - » **Operational models** has enabled efficient port monitoring.
 - » **Port business has become more competitive** with more terminals chasing users.
 - » **World Bank** too has praised India's PPP Model, saying the country witnessed considerable growth in PPPs in the last one and a half decade. India has emerged as one of the leading PPP markets in the world, due to several policy and institutional initiatives taken by central as well as many state governments.
- **Scale of PPP Projects:**
 - » The PPP investment numbers are quite huge in the port sector with nearly 300 PPP projects costing around 3.5 lakh crores taken up as of Dec 2019.
 - » The Sagarmala program has identified 123 PPP projects at an estimated investment of Rs 2.63 lakh crore.
 - Of this around 30 have been completed and around 30 additional are currently under implementation.
 - **Some successful examples:**
 - » **Mundra Port (BOT Model):** Adani Ports and Special Economic Zone (APSEZ) entered into a BOT agreement with Gujarat maritime board to develop and operate Mundra Port.
 - The port has emerged as one of the largest and most efficient ports in the country. It has seen a substantial cargo handling growth, established strong connectivity, and implemented state of art technology and infrastructure.
 - » **Kandla Port (Joint Venture):** Kandla International Container Terminal (KICT) was developed through a JV between ABG port and Deendayal Port Trust.
 - Kandla International Container Terminal (KICT) was developed through a JV between ABG Ports and the Deendayal Port Trust. This has significantly enhanced the container handling capacity at Kandla Port.
 - **Some challenges:**
 - » **Risk Allocation** between public and private sector.
 - » **Evolving Regulatory Framework:** The regulatory framework for PPPs in India is still evolving. This can create uncertainty for private investors and make it difficult to attract investment.
 - » **Lack of competition:** The huge size and complexity of PPP projects can discourage smaller contractors from bidding, so reducing competition, which may also effect the final cost.
 - **Conclusion:**
 - » These successful PPP projects demonstrate how private sector participation has positively impacted the efficiency, capacity, and overall performance of ports in India. However, it is important to be aware of the limitations of this approach and to take steps to mitigate the risks involved.

4) DIFFERENT TYPES OF PPP MODEL USED IN PORT SECTOR

- **Example Question:** Highlight the advantages and limitations of the different types of PPP models used in the development of ports in India [10 marks, 250 words]
- **Introduction:**
 - » Speak about need of PPP in Ports sector
- **Different types of PPP models used in Port development include:**
 - » **Landlord Port Model**
 - The landlord port model is a port management model in which the **port authority acts as a landlord, leasing the port infrastructure to private companies that operate the port.**
 - **The port authority** retains the ownership of the port infrastructure and retains the responsibility of the strategic port function, such as planning, regulation, and safety.
 - **The private companies** are responsible for the day-to-day operations of the port, such as cargo handling, stevedoring, and terminal management.
 - **Advantages:**
 - **Port Authorities** can focus on core competencies, while the private companies bring in their expertise in operations and management. This leads to a more efficient and effective port operations.
 - **Investment:** The model also attracts private investment, which can help to improve the port's infrastructure and facilities.
 - **Limitations:**
 - Private sector may not be willing to invest in the construction and operation of the superstructure if the revenue risk is very high.
 - **E.g.:**
 - **Jawaharlal Nehru Port** (India) became India's first 100% landlord port in July 2022.
 - » **Build Operate and Transfer (BOT) Model**
 - » **Build Own Operate Transfer (BOOT) Model**
 - Here, a private player is responsible for financing, constructing, owning and operating the port facility for a specified period before transferring it to the government.
 - **Advantages:** During concession period, private players have greater control over operations
 - **Limitations:** The private sector may not be willing to take on the risks associated with the project if the returns are not attractive.
 - » **Build Operate Own (BOO) Model**
 - The private sector builds a new port or terminal, operates it indefinitely, and own it outright.
 - Unlike BOT and BOOT, the private sector party owns the project and doesn't have to transfer it to the government entity at the end of the term.
 - **Advantages:**
 - Efficiency gains associated with private sector
 - Private sector has incentive to invest in long term infra.
 - **Limitations:**
 - Private sector may not be interested if there is no guarantee of long term return.

» **Management Contract Model**

- Private sector manages the port or terminal on behalf of the public sector. They bring their expertise and resources to bring efficient operation.

- **Advantages:**

1. Government benefits from private entity's expertise in port management
2. Reduced financial risks for the private entity compared to other models.

- **Limitations:**

1. Limited incentives for the private players to invest in infrastructure upgrade

- **Landlord port** model is the most common type of PPP model used in India. It is because it offers a good balance between risks and rewards for both the public and private sector.
- The choice of PPP model will depend on a number of factors, including the size and complexity of the project, the risks involved, and the objectives of the public and private sectors. It is important to carefully evaluate all these factors before selecting the PPP model.

5) COASTAL SHIPPING

- Advantages

- i. **Lowers logistic cost:** Government estimates that promoting coastal shipping of only six commodities, including coal, cement and steel, could result in a huge Rs 40,000 crore annual savings.
- ii. **Alleviate pressure on other modes** – especially roads and railways
- iii. **Environment friendly** -> fuel efficient, less pollution, less greenhouse gas emission
- iv. **Improve financial performance of ports and logistic companies too**
- v. **Development of coastal region**
- vi. **It also helps national security** as India's largely uninhabited coastlines leave them vulnerable to smugglers and terrorists.
- vii. **Promotion of cruise tourism** will be another benefit of promoting coastal shipping
- viii. **Reinvigorate ties between India and Bangladesh** and between India-Pakistan and India-Myanmar to some extent.
 - It can reduce the cost of goods transport to north-east.

- Key factors which have worked against India's Coastal Shipping Industry

1. **History:** Most coastal regions remained quite under-developed for centuries. Contrary to global experience where coast-based industries were the fastest to grow and emerge, India witnessed a stay away from coast approach. The Chola kings, who sent navies to South East Asia, are the only major examples of sea-faring power. In other words, the need for a navy was fully perceived after the British came to the shores.
2. **Geographical Factors:** East west shipments by sea, have to go round the southern tip of the country, leading to many companies preferring road and railways for the relative directness of speed and connectivity.
3. **India's regulatory environment has not helped historically** (Cabotage laws, restrictions on foreign flagged vessels, high domestic duties on oil, high tonnage taxes etc.)
4. **Coastal Regulation Zones** made it difficult to develop industry or commercial activities near the coast.

- Recent Steps to promote coastal shipping

- Mercifully, India has woken up from the Myopia. The new political will has emerged to promote coastal shipping. Recently government has taken a number of steps to promote coastal shipping:
 - a. **Sagarmala Project**
 - b. **Tweaking of Cabotage rules:** Now, specific, specialized foreign-flag vessels are allowed to operate along India's coast.
 - E.g. Cruise liners and Ro-Ro Boards have been allowed as India lacked its own cruise liners and Ro-Ro Boards.
 - c. Introduction of subsidy to encourage shippers to use coastal transportation
 - d. **Other steps**
 - Issuing Coastal Shipping Rules for coastal vessels operating within 20 miles off the coast.
 - Exempting Custom and Central Excise Duty on bunker fuels (IFO 180 and IFO 380CST) by Indian flagged coastal container vessels

6) SHIP BUILDING

- **Introduction:**
 - » Shipbuilding as an industry has deep historical roots in India.
- **Significance of Ship Building Industry:**
 - » **Reducing the cost of ships** -> Reduce cost of operations -> reduce freight bill and forest outgo and even earn foreign exchange.
 - » **Development of heavy engineering industry**: Ships would require various components from these industries.
 - » **Boost to ancillary industries**: The sector generates opportunities for smaller businesses and strengthens supply chain networks.
 - » **Growth and Employment**: Shipbuilding has one of the highest employment multiplier of 6.4 and will contribute to mass employment in remote, coastal and rural areas.
 - » **Expansion of blue economy**: Through offshore exploration, fisheries etc.
 - » **Role in Energy Security**: As most of the fossil fuel imported in India is through ships.
 - » **Defence**: Various Indian Navy Shipbuilding projects are posed to provide enhanced defence capabilities to India.
- **Current Situation in India:**
 - » India shipbuilding industry has continued to concentrate on defence, coastal and inland vessels.
- **Globally**, Shipbuilding industry is dominated by 3 countries namely, South Korea, China and Japan, which together have more than **90 percent** share of the shipbuilding market.
- **Key challenges faced by Shipbuilding sector in India:**
 - » **High Capital Cost.**
 - » **Lack of skilled workforce** - shortage of skilled engineers, designers and technicians.
 - » **Global competition** from giants of South Korea, Japan and China.
 - » **Policy and Regulatory challenges**
- **Key steps taken to promote shipbuilding in the country:**
 - » **Shipbuilding financial assistance policy scheme** was launched to offer financial support to Indian Shipyards for shipbuilding contracts signed between 2016 - 2026.
- **Way Forward:**
 - » Continue and enhance the Shipbuilding Financial Assistance Policy
 - » Provide tax benefits and concessions
 - » **Infra development** - Modernization of shipyards and improve port infrastructure to handle more advanced ships.
 - » **Focus on Human Resources**: More courses related to ship engineering at undergraduate and postgraduate levels.
 - » **International Collaboration** - Encourage joint venture, technology transfer etc from ship building giants of Japan, South Korea etc.
 - » **Simplify Regulations**: Reduce bureaucratic hurdles, make environmental clearance systematic
 - » **Focus on sustainability**: Adopt green technologies and encourage use of cleaner fuels.

7) SHIP RECYCLING

- About Ship Recycling

» Though, the ship recycling industry is a green industry as it allows recycle and reuse of many components of ship, at the same time, it creates a number of challenges in terms of environmental pollution, safety of workers etc.

» **India is a world leader** (after Bangladesh) in Ship Recycling sector with a market share of 26%. Bangladesh in recent year has emerged as the most important player in the industry as it pays higher value per kg of ship and also has abundant labor force.

- **Note:** Gujarat's Alang is the world's biggest shipyard and is expected to take more load in future. It is responsible for 98% of the total ships recycled in India. The other centers are Kolkata in WB and Azhikkal in Kerala.

- It is a labor intensive industry and thus also provide a potential to increase employment opportunities in the country. After the coming into force of the Recycling of Ships, Act, 2019 the Government of India is targeting a market share of 60%.

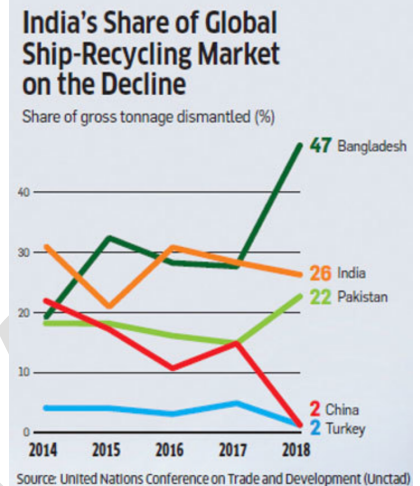
- Ship Recycling Act, 2019 (Came into force in 2020)

- **Need:** Ship recycling sector was poorly regulated which created negative environmental impact and hazardous condition for workers.
- The act regulates the ship recycling industry by setting standards, providing statutory mechanism for enforcement of the standards and restricting the use of hazardous material on ship.
- It also provides that ship recyclers should maintain adequate measures for emergency preparedness, response, safety, health, training, and welfare of workers as per the Factories Act, 1948. It must also provide insurance cover for regular and temporary workers.
- Under the act, government has notified Directorate General of Shipping as National Authority.

- India has also acceded to the Hong Kong International Convention (HKC) for Safe and Environmentally Sound Recycling of Ships, 2009 in Nov 2019.

- It is a convention under IMO which aims to ensure that recycling of ships don't pose any unnecessary risks to human health, safety and to the environment.
- It also addresses the working and environmental conditions at recycling facilities.
- **Note:** The convention will enter into force on 26th June 2025, after meeting the necessary ratification criteria.

- **Conclusion:** With ratification of HongKong Convention and passage of 2019 act, India's ship recycling industry will become more environment friendly and safe for workers. It will also increase business for ship recyclers by bringing in ships from USA and Japan which were not sending ships to India earlier because of non-ratification of the HongKong convention.



8) INLAND WATERWAYS

- **Example Questions**
 - » "Inland waterways in India may contribute immensely towards sustainable, inclusive and rapid economic growth" Elaborate. [10 marks, 150 words]
 - » What are the key factors hindering the development of Inland Water Transport in India? Discuss how a consultative, inter-disciplinary approach, as opposed to a techno-centric one, would keep the negative consequences of National Waterways development to its minimum [15 marks, 250 words]
- **Introduction**
 - » Inland Water Transport was an important means of trade and communication in ancient India. But, with the development of roadways and railways in 19th and 20th century, it got sidelined.
- **Inland Waterways Potential in India**
 - » Inland water transport holds great untapped potential as a means for the transportation of goods and passengers. India has a large endowment of rivers, canals, and other waterways. The total navigable length of waterways in India is around 14,850 kilometres
- **Need for improving and Inland waterways and ports**
 - i. **Contribution in trade** Only 3.5% of trade in India is done through the mode of Inland waterways, which is 47% in China, 40% in Europe, 44% in Japan and Korea and 35% in Bangladesh.
 - ii. **Fuel Efficient -> Cost Effective**
 - iii. **Less Polluting:** Environment friendly as the emission per unit distance per unit weight is lesser.
 - iv. **Economic growth and jobs:** It will boost economy and provide more jobs especially in inland regions which needs desperate boost to economies.
 - v. **Reducing Pressure on Road -> less congestion**
 - vi. **Fewer accidents** - when compared to any other mode of transport
 - vii. **Less Land Acquisition Problems and Less Deforestation:** As land capital required in case of water transport is minimal when compared to road and rail transport.
- **Challenges faced by Inland Waterways development in the country**
 - i. **Finance:** According to Ministry of Shipping, India needs Rs 80,000 crores to develop 20,000 km of inland waterways.
 - But, capital expenditure of Inland Waterways Authority of India (IWAI) for FY24 was just Rs 1010.5 crores.
 - ii. **Who will use inland waterways:** For e.g., there is not a lot of industries situated along the Ganga river.
 - iii. **Seasonal Variation in Navigable depth:** For e.g. around 20/111 waterways have been found unviable.
 - iv. **Most river transport routes are short:** River transport routes should run uninterrupted for at least 300 km at a stretch so that desired logistics cost saving might be achieved.
 - v. **Dams and Barrages interrupt the water supply:** Tussle between states would lead to unviability of some of these projects.
 - vi. **Environmental Concerns** (e.g. how EN Ganges river dolphins are being affected by NW-1)
- **Other Concerns** -> low transport speeds, limited areas of operation etc.

- **Key Initiatives to promote Inland Waterways:**

Inland Vessels Act, 2021 replaced the 1917 act and brought uniformity in legislation across states, standardization of classification of vessels, standard mechanism for registration of vessels etc to ease running of vessels across states. It also provided for higher standards regarding safety, navigation, environmental pollution, etc.

Inland Waterways Authority of India Act, 1985 -> Created a statutory authority for development and regulation of inland waterways.

National Waterways Act 2016

- **Commenced in 2016**
- **Provisions**
 - The act merged five erstwhile acts which had declared 5 National Waterways. It also proposed 106 additional National Waterways.
 - The act has thus declared 111 rivers or river stretches, creeks, estuaries as National (inland) Waterways (including the five older ones)
 - Now, according to entry 24 of the Union list of the seventh schedule, the union government can regulate these waterways for development with regard to shipping, navigation and transport through mechanically propelled vessels.

Other steps taken to promote National Waterways Recently

- i. Sagarmala Project
- ii. Implementation of **Jal Marg Vikas Project (JMVP)** to **augment capacity of NW-1** with the technical and financial support of the WB.
- iii. Ro-Ro and Ro-Pax have been initiated in some waterways.

- **Way forward**

- **More Funds allocation for Inland waterways:** Both government budgetary allocation and emphasis on enhancing private investment will be crucial here.
- **Increasing the demand of waterways:**
 - » Simultaneous focus on development of industries in the inland water ways region to ensure that waterways get used more and more and are economically viable.
 - » Improve connectivity of waterways to roads, railways, ports etc.
 - » **International Collaboration** - to ensure inland waterways can be used for international transportation also. For e.g. **India-Bangladesh Protocol (IBP) route**, developed jointly by India and Bangladesh provides an alternative connection for all NE states from Guwahati and Jogighopa to Kolkata and Haldia.
- **Dams and diversion of water for agriculture can be reduced** by focusing on newer dam technologies which limit the amount of water that needs to be stored and promoting efficient utilization of water in agriculture.
 - » In undertaking this paradigm shift towards inland transport, India should implement the sustainability practices in Europe.
- A consultative, inter-disciplinary approach, as opposed to a techno-centric one, would keep the negative consequences to a minimum.